

Study Companion:
The Statistics and Economics of
Pricing the DeFi Tail:
Do Protocols or Depositors Price Operational Risk?

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This document is a step-by-step companion to the main paper. Where the paper is terse, this document is verbose: it spells out the statistics, provides intuition for each modeling choice, and connects the derivations to standard results in extreme-value theory, actuarial risk measurement, and the economics of market discipline. Read this side-by-side with the paper.

1 The setup

The setup has four pieces: the economic object (a DeFi “earn” product), the risk being measured (operational risk, in the banking sense), the question (who bears the tail and is it priced), and the yardstick (a Basel loss-distribution approach). Each piece is a deliberate choice. This section walks through every choice with its motivation.

1.1 The economic object: an earn product

DeFi offers yield-bearing *earn products*: a lending-pool supply position, a decentralized-exchange (DEX) liquidity-provision position, a yield vault, a stablecoin LP position. Each pays a yield for supplying capital, and each is presented to users as a low-risk place to park value. I hold one reference point fixed throughout the paper, the *bank deposit*, and read every earn product against it. The introduction to the paper frames the comparison in three beats, and this subsection walks through each and then expands.

Similar in one respect: exposure to operational risk. The paper’s opening similarity claim is deliberately narrow. Similar to a bank deposit, an earn product exposes its holder to *operational risk*, the losses from failed processes, people, systems, and external events (Section 1.2 makes the Basel definition precise). In either product the depositor hands over capital and holds a redeemable claim, and in both an operational failure, a fraud, a system break, an external attack, can impair that claim. That shared exposure is the entire point of comparison. The paper does *not* claim the two products are otherwise alike, and neither do I.

Unlike a bank: no regulated capital behind the claim. The decisive contrast is what stands behind the claim when an operational loss occurs. A bank must hold operational-risk *capital*, sized under Basel from a *loss-distribution approach* (LDA, Section 3), so a regulated buffer sits between a loss event and the depositor’s claim. A DeFi protocol faces no such requirement, and by default holds nothing. The two products therefore expose the depositor to the same *kind* of risk while placing opposite amounts of capital behind the claim: alike in the exposure, opposite in the protection.

The decentralisation illusion. It is tempting to think an earn product is *safer* than a deposit, because DeFi’s architecture genuinely removes several risks that sit on a bank’s balance sheet. Smart-contract custody replaces the custodian, settlement is atomic, and counterparty credit risk collapses into the overcollateralization embedded in the contract. Aramonte et al. (2021) name the mistake the “decentralisation illusion.” The risks DeFi removes are custody, settlement, and counterparty credit. The risk it keeps, and in some cases concentrates, is *operational*: external attacks on contract code, insider-rugpull and internal-fraud events, oracle and economic-design manipulation, configuration and deployment errors. Removing the balance-sheet risks does nothing to the operational tail, and it is exactly that tail this paper measures.

The decisive difference: who bears the residual. Put the three beats together. In banking an operational loss hits the bank’s capital first, and the depositor is further insulated by deposit insurance and a resolution authority that can wind the bank down without touching an insured claim. DeFi has none of this: no capital requirement, no deposit insurance, no resolution authority. So the residual operational loss lands on the last claimant, the depositor, not the protocol. The risk is *transferred to the depositor by construction*. That transfer is the pivot of the paper, and Section 1.3 develops the two responses that can absorb it.

Why the bank deposit is still the right benchmark (the expansion). Given that the two are opposite in protection, why measure one against the other at all? Because the similarity that matters for a depositor is *perceived*. An earn product is marketed as a low-risk, yield-bearing place to park value, and its mechanics, deposit, accrue yield, redeem, mimic a deposit closely enough that an unsophisticated retail depositor, absent standardized risk disclosure, may take it to be deposit-like and price it as if the same protections stood behind it. The bank deposit is the benchmark the depositor *implicitly assumes*, so it is the right yardstick for asking whether the protection the depositor assumes is the protection that is actually there. The gap between the two is the subject of the whole paper.

Remark 1 (Perceived deposit, not an equivalent one). “Deposit-like” throughout describes how an earn product is marketed and perceived, not a claim that it carries the same risk as an insured, capitalized bank deposit. It does not. The comparison is worth drawing precisely because the perception and the protection diverge, and the size of that divergence is what the paper quantifies.

Why hold the methodology fixed and swap only the data? This is the paper’s central rhetorical move. If I measure the DeFi tail with *exactly* the machinery Basel uses for banks (the loss-distribution approach, Section 3), then any difference in the conclusion is attributable to DeFi, not to a different measuring stick. I do not pick a favorable method; I inherit the supervisor’s, so the comparison to a bank deposit is made in the bank supervisor’s own units.

1.2 The risk: operational risk in the Basel sense

Definition 2 (Operational risk). The risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events. It *excludes* market and credit risk.

Operational risk is the residual category — fraud, system failure, execution error, external attack. It is exactly the category DeFi cannot design away: smart contracts remove custodial and counterparty-credit risk, but the code can still be exploited, the team can still misconfigure an oracle, and the founder can still rug the pool.

The Basel Level-1 taxonomy. Basel II (BCBS d128, 2006) fixed a seven-category event-type taxonomy, carried into the Basel III finalization. I map five of them onto an autonomous protocol; the two I omit — Employment Practices and Damage to Physical Assets — have no on-chain analogue. The five that transfer:

Code	Basel Level-1	DeFi operational reading
IF	Internal Fraud	rugpull, owner-drain, insider exit-scam, malicious upgrade
EF	External Fraud	contract bugs (reentrancy, access-control, rounding), key phishing, DNS hijack
CPBP	Clients, Products & Business Practices	economic-design attacks: flashloan-governance, oracle/price manipulation, share-price inflation
BDSF	Business Disruption & System Failures	chain-level: MEV, sequencer outage, chain halt
EDPM	Execution, Delivery & Process Mgmt	team-side config/deployment errors: oracle mis-deployment, broken governance params

Why the taxonomy is orthogonal to the DeFi sector. The DeFi *sector* (Lending, DEX, Bridge, ...) is the product's *economic function*. The Basel type is the *failure mechanism*. They are different axes, so crossing them is genuinely informative rather than redundant. Empirically (paper Table 3) Bridge losses are almost pure EF (code drains pooled collateral), while Lending and Stablecoin losses are mostly CPBP (the economics were gamed, not the bytecode). Keep this orthogonality in mind: it is why the paper fits *per sector* rather than pooling (Section 3.7).

1.3 The question: who bears the tail, and is it priced?

Remark 3 (The pivot of the whole paper). In banking, a loss hits the bank's capital first, and the depositor is further insulated by deposit insurance and a resolution authority. In DeFi there is no insurance and no resolution authority, so the loss lands on the last claimant — the depositor. The residual risk is *transferred to the depositor by construction*.

Given that, only two things can absorb the tail, and the paper is organized around them as *competing responses by different participants*.

Response 1: the protocol holds a capital buffer. Some protocols keep governance-owned on-chain reserves that perform the function Basel calls capital: Aave's Umbrella safety module, the Sky (formerly MakerDAO) surplus buffer, Compound's per-market reserve factor. No regulation requires this, and the buffers that exist are sized by governance vote, not by any loss model.

Response 2: the depositor demands a risk premium in the yield. Where the protocol holds no buffer, the depositor should require a supply yield above the risk-free rate, sized to the tail they bear. This is the classic *market-discipline* mechanism by which

uninsured bank creditors price institutional risk into the yields they demand (Flannery 1998; Egan, Hortaçsu & Matvos 2017).

Why the two responses are substitutes. This is the conceptual key to Section 7. A buffer and a yield premium fund the *same* tail from two sides. If the protocol pre-funds with a buffer, the depositor needs less compensation; if it holds nothing, the depositor should demand more. Egan et al. (2017) prove exactly this for banks — deposit rates and capital are jointly determined substitutes. So an efficient DeFi market should show a *negative* relationship between buffer coverage and yield premium. That prediction is what the Mann–Whitney test in Section 7.3 checks.

1.4 The yardstick: a loss-distribution approach

The measurement tool is the standard operational-risk *loss-distribution approach* (LDA), the same one a bank supervisor applies under Basel. I develop it in full in Section 3. For now, note only its shape: it turns a list of event losses into an annual-loss distribution, and reads a capital number off that distribution’s tail. Because the tail of the annual loss is driven almost entirely by the tail of a *single* event’s severity, the whole construction stands or falls on estimating one number well — the tail index ξ (Section 4).

2 The dataset and the Basel tagging

A tail fit is only as good as its tail observations, so the data construction is methodologically central, not preliminary bookkeeping.

2.1 Why seven sources

Every prior DeFi/crypto-loss study leans on *one* feed, each with coverage gaps. Tail estimation is maximally sensitive to *missing large events*, so the paper merges seven public feeds: DefiLlama, rekt.news, DeFiHackLabs, kismpl23 DSI, BlockSec, de.fi/rekt-database (4,030 records), and SlowMist (2,100 records). The last two are broad and noisy — they include memecoin honeypots, CEX failures, individual phishing — which is why the filter below matters.

2.2 Step 1: normalize and deduplicate

The same event appears in several feeds under different names, so naive concatenation double-counts. I deduplicate in two passes.

Pass A — name clustering. Cluster records whose names match (fuzzily) within a 14-day window.

Pass B — date-and-amount. Match records on the same date (± 7 days) *and* loss within $\pm 10\%$. This second pass catches *cross-named* duplicates that Pass A misses — the canonical example is “Ronin Bridge” versus “Ronin Network,” the same USD 624 m event under two names.

Reconciling disagreeing amounts. When several sources report the same event with different dollar figures, the reconciled loss is the *median* of per-source amounts. Median, not mean, is deliberate: it is robust to a single feed’s outlier misreport. Because one inflated figure sits in the tail — exactly where the estimate is most sensitive — a mean would let a lone bad number distort ξ .

2.3 Step 2: filter to DeFi operational risk

Two filters keep only losses that are (a) on a DeFi *protocol* and (b) *operational*. A regex over event names removes CEX events, custodial-wallet failures, Ponzis, unattributed phishing, and wallet-software exploits; a sector-gated second pass drops NFT-mint, meme-coin, gambling, GameFi, and trading-bot records.

2.4 Step 3: infer sector and Basel type

Sector inference. Each event is assigned one of seven sub-sectors (Bridge, Lending, DEX, Yield, Stablecoin, Derivatives, Other). I use DefiLlama’s category where present, else a regex chain that resolves *Bridge first* (it is the most distinct and most consequential), then Lending, DEX, Stablecoin, Yield, Derivatives.

Risk-category (Basel) inference. Textual evidence (technique, description, name) is applied *first*, with DefiLlama’s generic classification only as a fallback, so an operational signature overrides a lazy label. The worked example is the February 2026 Moonwell cbETH event: an oracle priced collateral at USD 1.12 instead of $\approx$ USD 2,200, and liquidation bots seized USD 1.78 m within minutes. The root cause was an oracle *mis-deployment* — a team-side error — so it is tagged EDPM, not EF (which would be a code exploit).

Why inferential tagging is defensible, and its risk. Tagging by inference is a judgment layer and a possible source of bias. The defense is transparency (the rules are stated, the code is released) plus the orthogonality check of Table 3. The honest caveat — and a good referee question — is how sensitive the per-sector ξ estimates are to reclassifying borderline events.

2.5 Coverage and the final sample

Restricting to events dated between 2020-01-01 (when DeFi first reached material scale) and the 2026-05-29 data cutoff gives a *1,317-event consolidated sample* (USD 10.84 B) over 2020-02-11 to 2026-05-29. Two audit layers, now folded into the consolidation pipeline, refine this to the working sample used for every headline number:

1. A *seven-agent sector re-audit* (one parallel pass per sector, stored in `mistags.json`) reclassifies 277 mistagged events: OHM-fork reserve-currency losses are moved to Stablecoin, and memecoin/launchpad/gaming/NFT/CEX reassignments are marked `NOT_DEFI`. The resulting 153 non-DeFi records are dropped during consolidation, so they never enter the dataset — leaving a *1,164-event DeFi dataset*.
2. A *depositor-facing rule filter* (rules R1–R6 covering governance-token dilution, borrower over-liquidation, treasury drains, router-approval hijacks, frontend/DNS hijacks, and individual-wallet phishing), applied when the DeFi dataset is loaded, drops the 89 events whose losses did not hit the sector’s earn user (LP, supplier, holder).

The result is the working sample of *1,075 events* and *USD 9.45 B*. Roughly half (536 of 1,075) carry cross-source confirmation, and across those the median loss-amount disagreement is only 1.9%. “Gross” means before any funds returned — a conservative choice, since netting recoveries would only lower every downstream number.

Assumption 1 (Gross loss and reporting completeness). Losses are measured gross of funds later returned, and the feeds capture the events that drive the tail. Under-reported small events bias the body of the severity distribution downward but leave the large tail events, which set the capital figures, intact.

Assumption 1 makes the estimates conservative in a signed way. Netting recoveries (about 6% of the aggregate) would lower every downstream number, and any missed small events would only lighten the body. Neither direction can inflate the tail-driven capital figure, so the reported numbers are, if biased, biased low.

3 The loss-distribution approach

This is the statistical core. The LDA has three parts — severity, frequency, and their compound aggregation — and one summary statistic (a tail risk measure). I build each in turn, then assemble.

3.1 Statement

For each sector I model the annual aggregate operational loss as a random sum

$$S = \sum_{i=1}^N X_i, \quad N \sim \text{NB}(\hat{\lambda}, \hat{\alpha}), \quad X_i \stackrel{\text{iid}}{\sim} F_X, \quad (1)$$

where N (the number of events in a year) is drawn from a negative-binomial *frequency* model, each X_i (a single event’s loss) is drawn from a *severity* distribution F_X , and the two are independent. The capital number is the 99.9% quantile of S , and the pure premium is its mean:

$$\text{capital} = \text{VaR}_{99.9}(S), \quad \text{pure premium} = \mathbb{E}[S].$$

The random sum in (1) is the classical actuarial collective-risk model, and it rests on three structural conditions worth stating explicitly because everything downstream inherits them.

Assumption 2 (Compound structure). Within a sector, the single-event severities $X_1, X_2, \dots$ are independent and identically distributed draws from one severity law F_X ; the annual count N is independent of the severities; and events do not influence one another’s size or timing.

What Assumption 2 buys is tractability: the mean, variance, and tail of S then follow from F_X and the count law alone, without a joint model. What it costs is realism on two fronts I return to. Severities are not truly identical across protocols of very different size (Assumption 8), and events are not truly independent in time (Assumption 4). Both cost-directions are signed later, so the reader can see which way each pushes the capital number.

Remark 4 (Why the tail of S is really the tail of one X). Operational-loss severities are *subexponential* (Section 4.2): a large annual total is one catastrophic event, not an accumulation of moderate ones. So $\text{VaR}_{99.9}(S)$ is governed by the far tail of F_X , and estimating that tail well is the whole game. This is why I spend most of my effort on severity.

3.2 Step 1: severity by peaks-over-threshold

I treat individual gross losses as iid draws from F_X and model only its *upper tail* parametrically, splicing an empirical body to a generalized-Pareto tail.

Why model only the tail, and why a GPD? The Pickands–Balkema–de Haan theorem is the tail analogue of the central limit theorem. It says that for a very broad class of distributions, the excess over a high threshold u ,

$$Y = X - u \mid X > u,$$

converges as $u \rightarrow \infty$ to a *generalized Pareto distribution* (GPD), *regardless* of the original F_X . So the theory tells me the shape to fit above u — it is GPD — and I need not commit to a parametric family for the messy body, where data are dense and I simply resample empirically.

Assumption 3 (GPD tail). Above the selected threshold u , the exceedances $X - u \mid X > u$ are exactly generalized-Pareto. This is the finite- u version of the Pickands–Balkema–de Haan limit, and it is what licenses the parametric tail fit while leaving the body non-parametric.

Assumption 3 is not taken on faith. The mean-excess plot (Figure 1), the QQ plots (Figure 8), and the Kolmogorov–Smirnov distances all test it directly, and all three are consistent with it in every sector (Section 6). The residual risk is that u is finite, so the limit is an approximation; the threshold-stability scan (Figure 3) is the guard against picking a u where that approximation has not yet taken hold.

The GPD. With shape ξ and scale β , the excess distribution has density and survival function

$$g_{\xi,\beta}(y) = \frac{1}{\beta} \left(1 + \xi \frac{y}{\beta}\right)^{-(1/\xi+1)}, \quad \bar{G}_{\xi,\beta}(y) = \left(1 + \xi \frac{y}{\beta}\right)^{-1/\xi}, \quad y > 0.$$

The paper reports the density; the survival function is the more intuitive object for a tail. The two parameters carry very different information: β is a *scale* (the size of a typical excess, in USD m), while ξ is the *shape* — how fast the tail decays — and is the single most important number in the paper (Section 4).

Estimation: maximum likelihood. Given exceedances $y_1, \dots, y_{n_u}$, the GPD log-likelihood is

$$\ell(\xi, \beta) = -n_u \log \beta - \left(\frac{1}{\xi} + 1\right) \sum_{i=1}^{n_u} \log \left(1 + \xi \frac{y_i}{\beta}\right). \quad (2)$$

There is no closed-form maximizer, so I optimize numerically with *multi-start Nelder–Mead*. Nelder–Mead is a derivative-free simplex method, robust when the surface is awkward (GPD likelihoods can be flat or have boundary issues near $\xi = 0$); multi-start means I launch from several initial points and keep the best maximizer, because heavy-tailed likelihoods are notoriously multi-modal.

3.3 Step 2: choosing the threshold u

Choosing u is *the* practitioner problem in POT, and it is a bias–variance tradeoff.

The tradeoff, concretely. Set u too low and you admit body observations that are not yet GPD-distributed — that is *bias*. Set u too high and you keep too few exceedances n_u , so $\hat{\xi}$ swings wildly from sample noise — that is *variance*. The right u is the lowest one at which the GPD approximation has “kicked in.”

The plateau-stability rule. I scan the quantile grid $\mathcal{Q} = \{0.50, 0.55, \dots, 0.90\}$ and take the *highest* q at which $\hat{\xi}(q)$ has stopped moving — stable to within $\tau = 0.30$ as the threshold rises (the “plateau” in a threshold-stability plot). Two practitioner floors guard the exceedance count: $n_u \geq 20$ (Embrechts, Klüppelberg & Mikosch 1997) and $n_u/n \geq 10\%$ (DuMouchel’s 1983 “top 10%” rule). If no q satisfies the plateau, I fall back to the lowest eligible quantile. The point is a *rule-based, reproducible* threshold: hand-picking u is the classic way to engineer a tail estimate.

The mean-excess diagnostic. The visual companion is the mean-excess plot. For a GPD the mean-excess function is *linear* in u ,

$$e(u) = \mathbb{E}[X - u \mid X > u] = \frac{\beta + \xi(u - u_0)}{1 - \xi} \quad (\xi < 1),$$

with slope $\xi/(1 - \xi)$. So an upward-sloping straight mean-excess plot above some u is a signature of a GPD tail, and its slope already tells you the sign and rough size of ξ . Figure 1 plots $e(u)$ per sector on log-log axes. Every sector’s curve is monotonically increasing and near-linear across two decades of threshold, which is precisely the visual signature of a Pareto-type ($\xi > 0$) tail. Bridge sits an order of magnitude above the other curves, the first visible sign of the heavy tail the POT-GPD fit later pins at $\hat{\xi} = 1.74$. The diagnostic matters because it is model-free: it certifies a GPD-shaped tail before any GPD is fitted, so the later parametric fit is confirming a shape the data already show rather than imposing one.

The Hill cross-check. The Hill estimator is a second, semi-parametric tail-index estimator that uses only the top $k + 1$ order statistics and makes no threshold choice beyond k . It is an independent route to ξ , so agreement between the Hill plot and the POT-GPD fit is genuine corroboration rather than the same estimator twice. Figure 2 plots $\hat{\xi}_H(k)$ per sector against the two reference lines that organize the whole paper: $\xi = 1$ (the infinite-mean boundary) and $\xi = 0.5$ (the infinite-variance boundary). Bridge plateaus in the band $[1.2, 2.5]$, above the infinite-mean line, while every other sector plateaus in $[0.4, 1.0]$ below it. This is the same finite-mean-versus-infinite-mean split the POT-GPD fits deliver, reached by a different estimator, which is why the two diagnostics together make the split credible.

3.4 Step 3: quantifying uncertainty in $\hat{\xi}$

Each $\hat{\xi}$ carries a 1,000-replicate *parametric bootstrap* 95% CI. “Parametric” means I simulate fresh datasets *from the fitted GPD*, refit each, and read the 2.5/97.5 percentiles of the refit $\hat{\xi}$ ’s. This propagates sampling uncertainty honestly, and it is what lets the paper say the Bridge CI $[0.79, 2.74]$ *straddles* the infinite-mean boundary: the point estimate exceeds 1, but the data cannot rule out a finite mean.

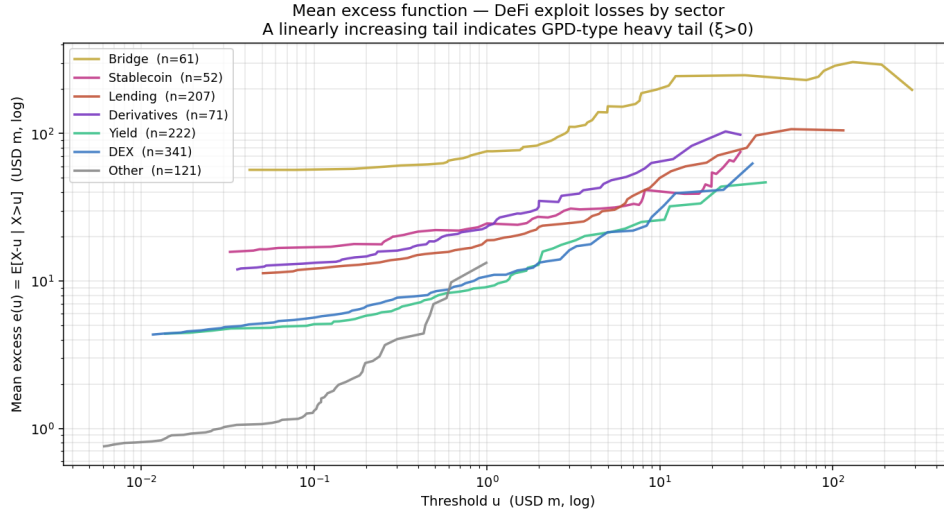


Figure 1: Mean-excess function $e(u) = \mathbb{E}[X - u \mid X > u]$ per sector (log-log axes). An approximately linear, upward-sloping curve is the signature of a generalized-Pareto ($\xi > 0$) tail. All seven sectors satisfy it across two decades of threshold; Bridge is offset upward by its six nine-figure mega-events, foreshadowing the heaviest $\hat{\xi}$.

The threshold-stability scan. The plateau-stability rule is easiest to see as a picture. Figure 3 plots $\hat{\xi}(q)$ against the threshold quantile q on a fine grid, with the bootstrap CI and interquartile range shaded at every q . The rule fires where the curve flattens into a plateau, and the picture shows why the guard floors matter. In the Bridge panel the estimate is stable from $q = 0.50$ to $q = 0.85$ and then collapses at $q = 0.90$, where only ten exceedances remain and the GPD MLE flips into Weibull-bounded territory. That collapse is small-sample noise, not signal, and the $n_u \geq 20$ and $n_u/n \geq 10\%$ floors are exactly what exclude it. Reading the plot is the honest way to confirm that each headline $\hat{\xi}$ comes from a genuine plateau and not from a lucky single threshold.

3.5 Step 4: frequency by negative binomial

For each sector I fit a count model to the *monthly* event series, $N_s \sim \text{NB}(\mu_s, \alpha_s)$.

Why not Poisson? The natural first model is Poisson, whose defining property is $\text{Var}(N) = \mathbb{E}(N)$: events arrive independently at a constant rate. Real exploit counts *cluster* — a new attack technique spawns copycats, a market crash triggers correlated liquidations — so the variance exceeds the mean. This is *over-dispersion*, and Poisson cannot represent it. The clustering is visible directly in the rolling event intensity (Figure 4): each sector’s 90-day event rate rises and falls in waves rather than fluctuating around a flat line, and the sector mix rotates over time (the early rugpull wave in 2021, Bridge mega-events in 2022, a broad lift in 2023). Two facts for the model follow from the same picture. The waviness is the over-dispersion the negative binomial captures, and the sector-mix rotation is the empirical warrant for fitting frequency *per sector* rather than pooling.

Assumption 4 (Event independence). Conditional on the fitted frequency law, events arrive independently: an exploit at one protocol does not raise the near-term hazard at

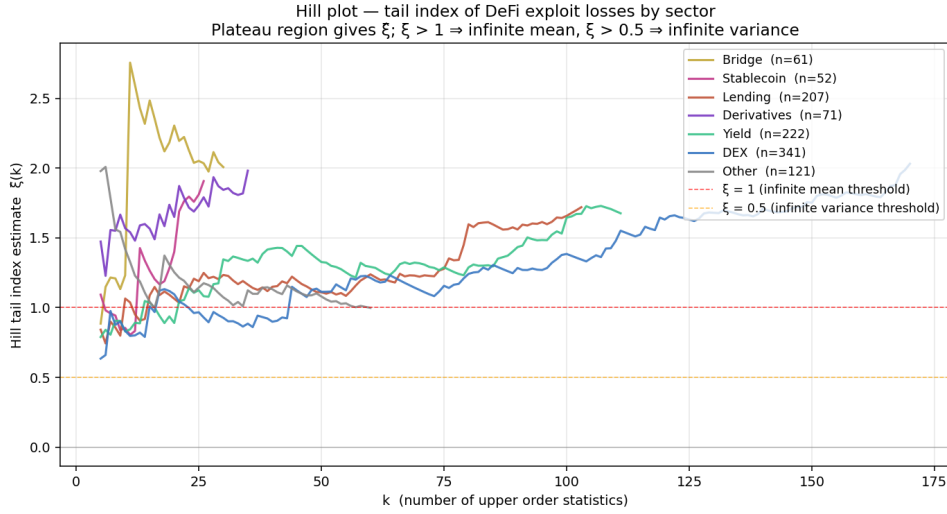


Figure 2: Hill tail-index estimate $\hat{\xi}_H(k)$ per sector against the number of upper order statistics k . A stable plateau in k is the usable estimate. Bridge sits above the infinite-mean line $\xi = 1$ (red dashed); the remaining sectors sit below it, between the infinite-variance line $\xi = 0.5$ (orange dashed) and 1. The split reproduces the POT-GPD ordering by an independent estimator.

another. The compound model draws the annual count once and the severities independently, with no self-excitation.

Assumption 4 is the one the paper is most explicit about violating. The wavy rolling intensity in Figure 4 is exactly the cross-protocol contagion a self-exciting (Hawkes) point process models, and the negative binomial captures the resulting over-dispersion only in the *marginal* count, not the *timing* dependence. The direction of the error is known. Ignoring positive contagion understates the probability of a cluster of large events in one year, so the reported $\text{VaR}_{99.9}$ is, if anything, a lower bound on the true tail. A joint Hawkes model is the natural extension (Section 8).

Fitting the whole window as one process. The same rolling-intensity picture that shows clustering also shows *trend*: the sector mix rotates and the overall level drifts across 2020–2026, so the event rate is not constant over the window. I nonetheless fit one frequency law (and one severity law) per sector on the full window.

Assumption 5 (Stationarity). The per-sector severity and frequency laws are constant over the 2020–2026 window, so a full-sample fit estimates the process that generates next year’s loss.

Assumption 5 is visibly an idealization, in two directions the same figure displays. Frequency fell from the early rugpull wave, and severity moved toward larger, more concentrated targets. The paper’s out-of-sample backtest quantifies the cost, and the direction of the error runs opposite to the independence one, so Section 8 takes the two up together with the fix for each.

The negative binomial as a Gamma-mixed Poisson. Let the rate itself be random: $\Lambda \sim \text{Gamma}$, and $N \mid \Lambda \sim \text{Poisson}(\Lambda)$. Marginalizing over Λ gives a negative binomial

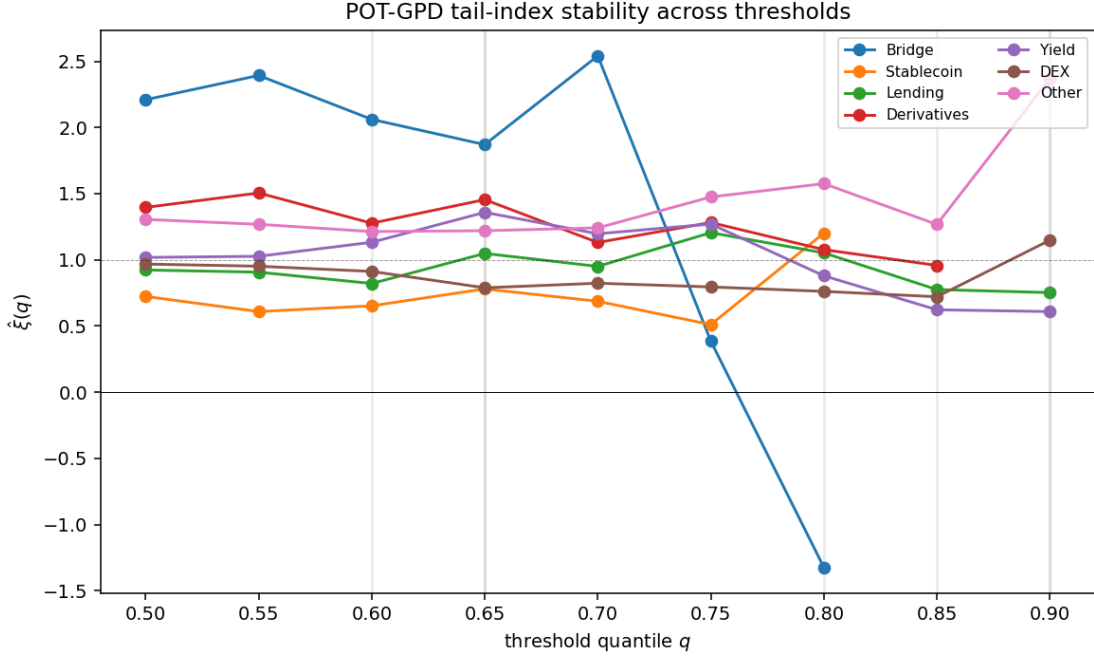


Figure 3: Per-sector parameter-stability scan: $\hat{\xi}(q)$ as a function of the POT threshold quantile $q \in [0.50, 0.90]$, step 0.05. Number annotations are the exceedance count n_u at each q ; red-ringed points flag $n_u < 15$, where the MLE asymptotics fail and the fit is unreliable. Shaded bands are the 1,000-replicate parametric bootstrap 95% CI (light) and IQR (dark). A flat region is a usable plateau; the Bridge collapse at $q = 0.90$ ($n_u = 10$) is the small-sample instability the count floors are designed to exclude.

with

$$\mathbb{E}(N) = \mu, \quad \text{Var}(N) = \mu + \alpha\mu^2, \quad (3)$$

where $\alpha \geq 0$ is the *dispersion*. At $\alpha = 0$ the variance equals the mean and I recover Poisson; $\alpha > 0$ is over-dispersion, and larger α means more clustering. Under-modeling dispersion would understate how many events a bad year can hold, and hence understate $\text{VaR}_{99.9}$ — you would report too little capital.

From monthly to annual: why the dispersion divides by 12. The compound (1) needs an *annual* count, which I take as the sum of twelve iid monthly draws. The paper states $\hat{\alpha}_{\text{annual}} = \hat{\alpha}_{\text{monthly}}/12$ in one clause; here is the derivation. Using (3), twelve independent months give

$$\mathbb{E}[N_{\text{yr}}] = 12\mu, \quad \text{Var}(N_{\text{yr}}) = 12(\mu + \alpha\mu^2) = 12\mu + 12\alpha\mu^2.$$

Now require the annual count to be NB with mean $\mu' = 12\mu$ and dispersion α' , so $\text{Var} = \mu' + \alpha'\mu'^2 = 12\mu + \alpha'(144\mu^2)$. Matching the two variance expressions,

$$12\alpha\mu^2 = 144\alpha'\mu^2 \implies \alpha' = \frac{\alpha}{12}.$$

Intuition: aggregating independent periods averages out clustering, so a year is proportionally *less* over-dispersed than a single month.

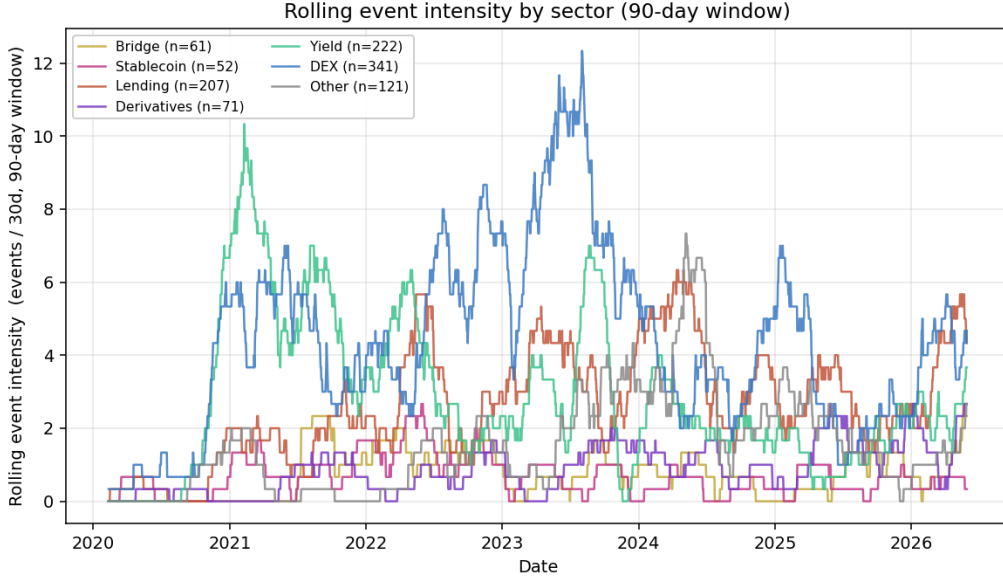


Figure 4: Rolling 90-day event intensity by sector (events per 30-day equivalent). Intensity moves in waves rather than around a constant rate, the temporal clustering that over-disperses the monthly counts and rejects Poisson. The relative sector composition also rotates over time, the descriptive warrant for the per-sector treatment.

The Poisson-versus-NB picture. Figure 5 makes the over-dispersion visible in the count distribution itself. The upper panel is the monthly event count over the sample; the lower panel overlays the fitted Poisson and negative-binomial probability mass functions on the observed histogram. The Poisson curve is too narrow: it concentrates mass near its mean and assigns almost no probability to the busy months of 30-plus events that actually occur. The negative binomial spreads mass into exactly that right tail. Because the compound $\text{VaR}_{99.9}$ is driven by high-count years, the gap between the two curves in the right tail is the gap between an adequate and an inadequate capital number.

3.6 Step 5: compound aggregation by Monte Carlo

There is no closed form for a compound NB-GPD, so I simulate 2×10^5 years per sector. Each simulated year: draw N from the annual NB, draw N severities from the empirical-body/GPD-tail mixture, and sum. From the 200,000 totals I read the mean $\mathbb{E}[S]$ (the *pure premium*) and the 0.999 quantile $\text{VaR}_{99.9}$ (the *capital buffer*).

The exposure cap. Each simulated single-event loss is capped at the largest single-protocol exposure in the sector, because one operational event cannot drain more than the assets at risk in one protocol. This is not cosmetic. For the finite-mean sectors it merely bounds the extreme quantile; for the $\xi > 1$ sectors it is the *only* thing that makes VaR finite at all. The paper reports both the capped figure and the uncapped $\text{VaR}_{99.9}^U$, and the uncapped Bridge value diverges to $286 \times \text{TVL}$ — a direct display of the infinite-mean pathology (Section 4).

Assumption 6 (Exposure cap). A single operational event cannot lose more than the assets at risk in one protocol, so each simulated loss is truncated at the largest single-protocol exposure in the sector.

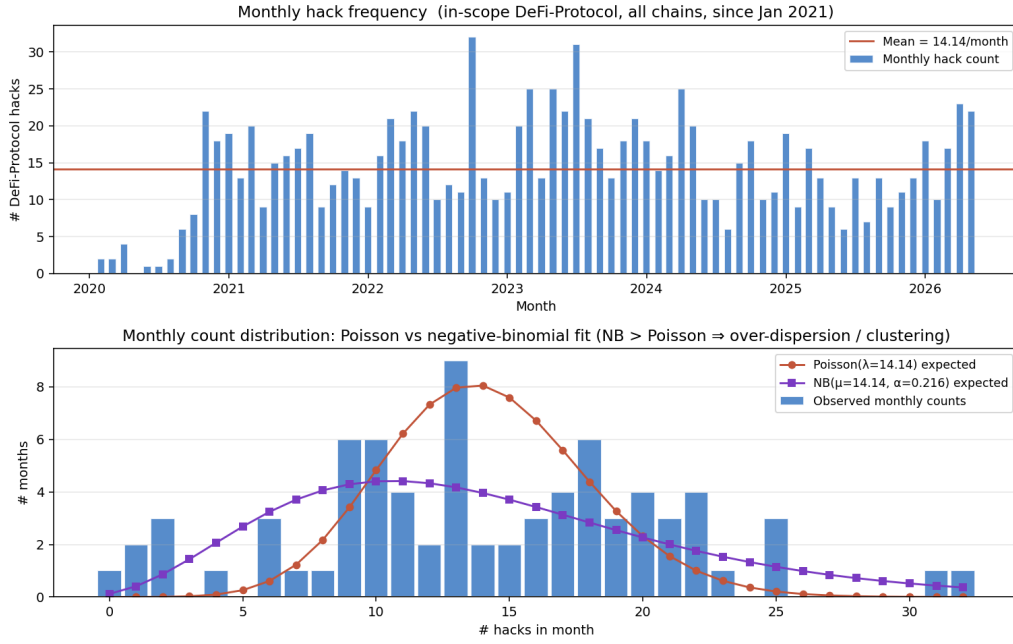


Figure 5: Monthly in-scope event counts (upper) and the cross-sector count distribution with fitted Poisson and negative-binomial PMFs (lower). The negative binomial ($\alpha \approx 0.2$) captures the right tail of busy months that Poisson under-counts. Because the compound $\text{VaR}_{99.9}$ is dominated by high-count years, that right-tail mass is what a Poisson model would leave out of the capital figure.

Assumption 6 is economically unimpeachable for a single event, but it does real work in the results and the reader should see where. For the finite-mean sectors it changes little. For Bridge and Derivatives ($\xi > 1$) it is load-bearing: without it there is no finite quantile at all, so the reported Bridge and Derivatives capital figures are properties of the cap, not of the fitted severity. The paper is explicit that these two rows are cap-determined floors rather than estimates, which is why it declines to read them as capital targets.

Expressing capital as a fraction of TVL. I report $\mathbb{E}[S]$ and $\text{VaR}_{99.9}$ as a share of each sector’s trailing-365-day total value locked (TVL). The share is the meaningful unit — it answers “what fraction of depositors’ money” — and it respects the cap, since a protocol cannot lose more than users supplied. One refinement matters and is worth making explicit once. The TVL used as the denominator is the *net* figure, the idle assets actually sitting in the protocol, not the gross supplied total. For a lending market, gross supplied TVL counts idle liquidity *plus* funds that have been borrowed out, and the borrowed portion has left the contract, so a drain-the-pool operational event cannot take it. Net TVL strips the borrowed leg back out and leaves the value genuinely exposed to an operational loss. Using net TVL also removes the recursive double-counting that inflates gross figures, where the same dollar is supplied, borrowed, and re-supplied and would otherwise be counted more than once. The paper does not repeat the word “net” on every mention of TVL, but every capital number (the severity cap, $\mathbb{E}[S]$, and $\text{VaR}_{99.9}$) is computed on this net basis. The one place gross supplied TVL is used is selecting which ten venues enter the sample, so that a high-utilization money market is not understated at the selection step. Two sectors lack a clean supply-side TVL, so I substitute the closest exposure base: decentralized-stablecoin circulating supply for Stablecoin, and residual

TVL for Other. Both are approximate and flagged as such.

Assumption 7 (TVL as the exposure base). Trailing-365-day *net* TVL (idle assets held in the protocol, gross supply less funds borrowed out) is *taken as* the exposure base for capital and the weight for allocating a sector loss to a protocol, because it is the pool of depositor funds an operational event can actually reach and it keeps the capital ratio consistent with the exposure cap (Assumption 6). Where a clean supply-side net TVL does not exist, the closest available exposure base substitutes (decentralized-stablecoin circulating supply for Stablecoin, residual TVL for Other).

Assumption 7 is deliberately hedged. Net TVL is the *appropriate* at-risk base, not a provably unique one, which is why the statement says “taken as” rather than asserting correctness. It is clean for the supply-side sectors (Lending, DEX, Yield) where net TVL is well defined, and admittedly approximate for Stablecoin and Other, whose substitute bases are flagged and kept out of the adequacy tests. It is also where value-at-risk can depart from locked value: gross TVL can still be inflated by mercenary, incentive-chasing deposits, and a larger venue may be better audited so that risk per dollar falls with size, the same concern the per-sector homogeneity assumption (Assumption 8) carries.

3.7 Why I fit per sector, not once

I fit severity and frequency *once per sector* rather than on the pooled data. The reason is the orthogonality of Section 1.2: the dominant loss mechanism differs by sector (Bridge is EF on pooled collateral; Lending and Stablecoin are CPBP economic-design failures). A single pooled fit would blend these distinct loss-generating processes and misestimate every tail. The cost is sample size — only 60 Derivatives events — but the heterogeneity it preserves is exactly what drives the per-sector capital numbers.

4 Reading the tail index ξ

Everything downstream — whether the mean exists, whether VaR is finite, how capital scales with venue size — is governed by where ξ sits. This short section is the conceptual heart of the paper.

4.1 The moment rule and two thresholds

Proposition 5 (Finiteness of GPD moments). *For a GPD with shape $\xi > 0$, the moment $\mathbb{E}[X^k]$ is finite if and only if $k < 1/\xi$. In particular the mean is finite iff $\xi < 1$, and the variance is finite iff $\xi < 1/2$.*

So two thresholds matter:

ξ range	Tail behavior	Finite moments
$0 < \xi < \frac{1}{2}$	heavy, finite variance	mean and variance
$\frac{1}{2} \leq \xi < 1$	heavy, <i>infinite</i> variance	mean only
$\xi = 1$	infinite-mean boundary	none (mean diverges)
$\xi > 1$	infinite mean	none

Why $\xi > 1$ is a crisis, not merely “heavy.” If the mean is infinite, the sample average is not a consistent estimator of anything: add more data and it keeps drifting up, dragged by the next record event. A capital number built on such a severity *diverges* — there is no stable 1-in-1000 loss. This is precisely the instability that led the Basel Committee to retire the internal-model Advanced Measurement Approach in 2017. The paper’s Bridge ($\hat{\xi} = 1.74$) and Derivatives ($\hat{\xi} = 1.16$) reproduce it: their point estimates cross the boundary, and only the exposure cap gives them a finite VaR. The paper is careful to add that both confidence intervals straddle 1, so the crossing is a point estimate, not an inference.

4.2 Subexponentiality and the single big loss

Definition 6 (Subexponential). F_X is subexponential if $\mathbb{P}(X_1 + \dots + X_n > x) \sim \mathbb{P}(\max_i X_i > x)$ as $x \rightarrow \infty$: a large sum is dominated by its largest term.

All the paper’s sectors are subexponential — the diagnostics (Section 6) confirm either a power-law or a lognormal tail, and both are subexponential. This property is what makes “a bad year is one bad event” literally true, and it is what licenses the single-loss approximation for per-protocol capital in Section 5.

4.3 Tail index versus power-law exponent

A GPD tail *is* a power law: $\mathbb{P}(X > x) \sim x^{-1/\xi}$, so the CCDF exponent is $1/\xi$. The paper also reports a Clauset power-law fit $\hat{\alpha}_{\text{PL}}$, but Clauset’s convention is the exponent of the *density*, $p(x) \sim x^{-\alpha}$, so the CCDF exponent there is $\alpha - 1$. Reconciling the two conventions,

$$\xi \approx \frac{1}{\alpha_{\text{PL}} - 1}. \quad (4)$$

This is a useful order-of-magnitude cross-check between two independent fits: Bridge’s $\hat{\alpha}_{\text{PL}} = 1.43$ implies $\xi \approx 1/(1.43 - 1) = 2.33$ against a fitted 1.87, and DEX’s 1.90 implies $1/(1.90 - 1) = 1.11$ against 0.72. The two fits agree on the heavy-versus-light ordering, with the Hill-type $\hat{\alpha}_{\text{PL}}$ (evaluated at u^*) running somewhat heavier than the GPD MLE.

5 Per-protocol allocation

The adequacy tests in Section 7 compare a *specific venue’s* buffer or yield against *its own* modeled capital. But no single protocol records enough events for its own fit. Bridging that gap is a subtle and important step.

5.1 The credibility-theory framing

When an individual unit has too little data for its own estimate, actuarial *credibility theory* (Bühlmann 1967; Bühlmann–Gisler 2005) prescribes how much weight to place on that thin individual signal versus a broader *collective*. This is exactly my position. A single protocol records a handful of events at most, far too few for its own tail fit, while its sector pools hundreds. Credibility theory is the principled bridge between the two, and it is worth spelling out because it is what licenses the whole per-protocol allocation.

The credibility estimator. Let each protocol p carry an unknown risk parameter Θ_p , and let $\mu(\Theta_p) = \mathbb{E}[X \mid \Theta_p]$ be its true, unobserved loss rate. The credibility estimate of $\mu(\Theta_p)$ is a convex combination of the protocol’s own experience and the collective mean,

$$\hat{\mu}_p = Z_p \bar{X}_p + (1 - Z_p) \mu_0, \quad (5)$$

where $\bar{X}_p$ is protocol p ’s own average loss, μ_0 the sector (collective) mean, and $Z_p \in [0, 1]$ the *credibility factor*. Equation (5) is the best linear approximation to the Bayesian posterior mean $\mathbb{E}[\mu(\Theta_p) \mid \text{data}]$, which is why it is the standard actuarial answer to “trust the unit or trust the pool.” At $Z_p = 1$ I use the protocol’s own number; at $Z_p = 0$ I hand it the sector’s.

What sets the weight Z . In the Bühlmann–Straub model, where each unit carries an exposure (volume) measure w_p , the credibility factor is

$$Z_p = \frac{w_p}{w_p + k}, \quad k = \frac{\sigma^2}{\tau^2} = \frac{\mathbb{E}[\text{Var}(X \mid \Theta)]}{\text{Var}(\mathbb{E}[X \mid \Theta])}. \quad (6)$$

The constant k is the ratio of the *expected process variance* σ^2 (the noise *within* a unit’s own history) to the *variance of the hypothetical means* τ^2 (the genuine heterogeneity *between* units). The entire content of credibility theory sits in this one fraction. The weight $Z_p \rightarrow 1$, so I trust the unit, when its exposure w_p is large or when units are truly different (τ^2 large, hence k small). The weight $Z_p \rightarrow 0$, so I trust the pool, when the unit is data-poor or when units are essentially alike (τ^2 small, hence k large). Credibility rises with own data and with real between-unit heterogeneity, and falls when either is missing.

Where this paper sits: near-total shrinkage. Two features of DeFi drive me to the $Z_p \rightarrow 0$ corner. First, per-protocol event counts are tiny, so $\bar{X}_p$ is almost pure noise and carries little weight in its own right. Second, I do not attempt to estimate the between-protocol heterogeneity τ^2 at all, which is the same as treating $\tau^2 \rightarrow 0$, so $k \rightarrow \infty$ and $Z_p \rightarrow 0$ regardless of exposure. Both send the weight to zero, and (5) collapses to $\hat{\mu}_p = \mu_0$: each protocol inherits the *sector* estimate outright. The natural volume measure w_p in (6) is net TVL, so “inherit the sector estimate” means “inherit it per unit of TVL,” which is exactly the TVL-share thinning of the next subsection, $\lambda_p = (\text{TVL}_p / \text{TVL}_s) \lambda_s$ with the shared severity. The per-protocol allocation is thus a credibility estimate pinned at zero own-credibility.

Assumption 8 (Per-sector homogeneity). All protocols in a sector share one severity law and one per-TVL event rate. A sector event lands on protocol p with probability proportional to its TVL share $\text{TVL}_p / \text{TVL}_s$, and conditional on being struck, p ’s loss is drawn from the same F_X (capped at its own exposure).

Assumption 8 is the price of thin per-protocol data, and it is the one a referee will press hardest. It says risk-per-dollar is constant within a sector, which a larger or more complex venue may violate by carrying more risk per dollar of TVL. The paper flags this as the *homogeneity* limitation and names the fix: a size covariate or a hierarchical (random-effects) severity model that lets each protocol’s tail depart from the sector collective as its own data accumulate. Absent that data today, the TVL-share allocation is the disciplined default, and the credibility framing is precisely the statement that it is a shrinkage estimate, not a claim of true equality.

5.2 Thinning the sector process to a protocol

Model the sector's event stream and *thin* it to a protocol by TVL share. Let TVL_p be protocol p 's TVL and $\text{TVL}_s = \sum_{p \in s} \text{TVL}_p$ the sector total. A sector event strikes protocol p with probability $\text{TVL}_p/\text{TVL}_s$, so p inherits a thinned rate

$$\lambda_p = \frac{\text{TVL}_p}{\text{TVL}_s} \lambda_s,$$

the same per-event severity, capped at its own exposure.

The mean thins linearly. Because expectation is linear, the pure premium allocates cleanly by TVL share:

$$\mathbb{E}[S]_p = \frac{\text{TVL}_p}{\text{TVL}_s} \mathbb{E}[S]_s.$$

So if I only cared about the average, per-protocol numbers would be a trivial rescaling. I care about the tail, and the tail does *not* thin linearly.

5.3 Why the tail scales as TVL_p^ξ

For subexponential severity, the *single-loss approximation* (Böcker–Klüppelberg 2005; Embrechts et al. 1997) says a high-quantile aggregate VaR is essentially the matching severity quantile:

$$\text{VaR}_{0.999}(S_p) \approx F_X^{-1}\left(1 - \frac{0.001}{\lambda_p}\right). \quad (7)$$

Now use the GPD quantile, which for a small tail probability pr behaves like $F_X^{-1}(1-pr) \propto pr^{-\xi}$. Substituting $pr = 0.001/\lambda_p \propto 1/\text{TVL}_p$,

$$\text{VaR}_{0.999}(S_p) \propto \left(\frac{1}{\text{TVL}_p}\right)^{-\xi} = \text{TVL}_p^\xi.$$

Therefore the *VaR-to-TVL ratio* scales as $\text{TVL}_p^{\xi-1}$. With $\xi < 1$ the exponent is negative, so **the smaller the venue, the larger its capital as a share of its own TVL**. Intuitively, a small venue is struck rarely, but when it is struck the loss is drawn from the same heavy severity law, so relative to its size a bad year is proportionally more devastating.

Remark 7 (Why I re-simulate rather than rescale). Equation (7) is why I cannot simply scale the sector VaR-ratio down to each protocol — that would understate the small venues. I instead re-simulate $\text{VaR}_{99.9}$ per protocol. This is exactly why, in the yield table (Section 7), the per-protocol $\text{VaR}_{99.9}$ premia (5,998–9,666 bps) dwarf the sector ratio ($\sim 1,800$ bps).

6 Reading the empirical fits

A guided tour of the paper's tables and figures, and what each licenses.

Severity fits (Table 4). The per-sector $\hat{\xi}$, read against the two thresholds of Section 4 and the Moscadelli (2004) banking band $[0.85, 1.39]$:

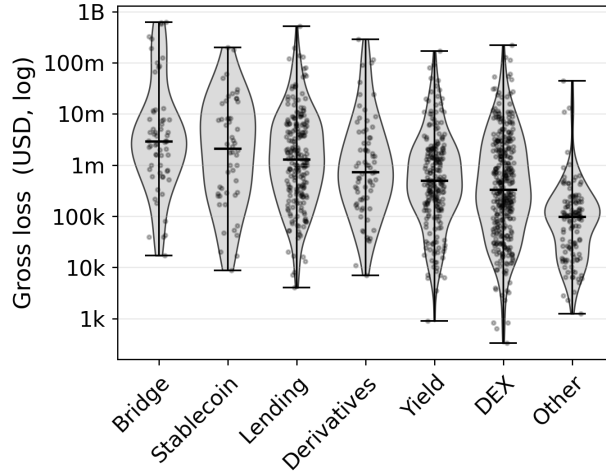


Figure 6: Per-sector loss distribution ($n = 1,075$; violin = density, bar = median), sectors by descending median. The violin reads the body of the severity distribution and shows the sectors do not overlay in location or shape, so a single pooled severity would blend genuinely different distributions. Bridge carries the highest median event loss, from the large pooled collateral cross-chain bridges hold.

Sector	$\hat{\xi}$	Reading
Bridge	1.87	past infinite-mean boundary; CI [0.58, 3.18] straddles it; cap-dependent
Other	1.58	past boundary; residual bucket, thin fit; cap-dependent
Derivatives	1.45	past boundary; only 71 events; unstable
Lending	0.75	heavy, finite mean; sensitive to Vires Finance (single event)
DEX	0.72	heavy, finite mean; carried by many events
Stablecoin	0.65	moderately heavy, finite mean
Yield	0.61	moderately heavy, finite mean

The understated headline: *the four finite-mean DeFi sectors are no heavier-tailed than banks*. The drama is not exotic tails but ordinary-heavy tails combined with no capital.

Cross-tabulation (Table 3) and severity figures. The sector $\times$ Basel table shows Bridge $\times$ EF alone is 33% of all losses, and confirms that mechanisms differ by sector — the empirical warrant for per-sector fitting (Section 3.7). The two severity figures make the same point visually. The violin plot (Figure 6) shows that the per-sector loss distributions do not overlay: they differ in both location and shape, so a single pooled severity would blend genuinely different distributions. Bridge carries the highest median event loss, from the large pooled collateral cross-chain bridges hold. The log-log CCDF (Figure 7) shows approximately straight tails, and a straight line on log-log axes *is* a power law, with slope the tail exponent. The two plots are complementary: the violin reads the body and the CCDF reads the tail, and both refuse to collapse the sectors onto one curve.

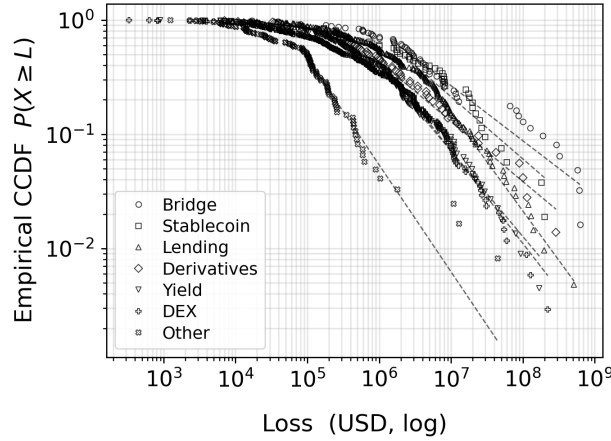


Figure 7: Log-log complementary CDF $\Pr(X \geq L)$ per sector ($n = 1,075$) with the Clauset power-law tail (dashed) overlaid. The CCDF reads the tail and shows approximately straight-line behavior over three decades, and a straight line on log-log axes *is* a power law with slope the tail exponent. Together with the violin (Figure 6) it is the descriptive warrant for a per-sector, heavy-tailed severity model.

Frequency fits (Table 5). Annual rates run from 8.3/yr (Stablecoin) to 54.2/yr (DEX). The likelihood-ratio column rejects Poisson in favor of NB in every sector but the two smallest (Stablecoin, $p = 0.59$; Derivatives, $p = 0.05$), and a cross-sector Cameron–Trivedi test gives an overwhelming dispersion index $D = 3.27$ ($z \approx 14$).

Per-sector capital (Table 6). The yardstick itself, in % of TVL, reported both capped and uncapped ($\text{VaR}_{99.9}^U$). Lending 18.0%, Stablecoin 16.1%, Yield 15.3%, DEX 14.1% — all far above the operational-risk capital banks hold. Bridge 32.5%, Other 27.5%, and Derivatives 22.8% are larger but *cap-determined*: uncapped, Bridge diverges to $673 \times$ TVL.

Goodness of fit, and what each test assures. A capital number is only as trustworthy as the fit beneath it, so the paper runs four separate goodness-of-fit checks, each closing a different failure mode. (i) *The QQ panel* (Figure 8) plots empirical excesses against the fitted-GPD quantiles per sector. A near-straight 45° line means the fitted tail matches the data at every quantile, not just on average, which is the check that matters for a 99.9% number. The panel is essentially straight in every sector, with a slight upper-tail over-shoot in Lending (Vires Finance, Euler V1) and Bridge (Ronin, Poly) where the data are very slightly *heavier* than the fit, so the reported figures there are, if anything, mildly conservative. (ii) *Kolmogorov–Smirnov distances* ≤ 0.19 (within the 5% critical band) give a formal, single-number version of the same statement: the fitted and empirical tails are not statistically distinguishable. (iii) *The Vuong power-law-versus-lognormal test* is near-tied for the lighter-tailed core sectors and favors the power law for the heavy-tailed ones (Bridge, DEX, Other; $\text{LR}_{\text{LN}} > 0$). This does not unsettle the capital story: power-law and lognormal are both subexponential (Section 4.2), so the 99.9% figure is identical either way, and where the test does discriminate it points to the heavier power-law tail, consistent with the infinite-mean point estimates in those sectors. (iv) *The frequency side* is checked by the per-sector likelihood-ratio test (rejects Poisson

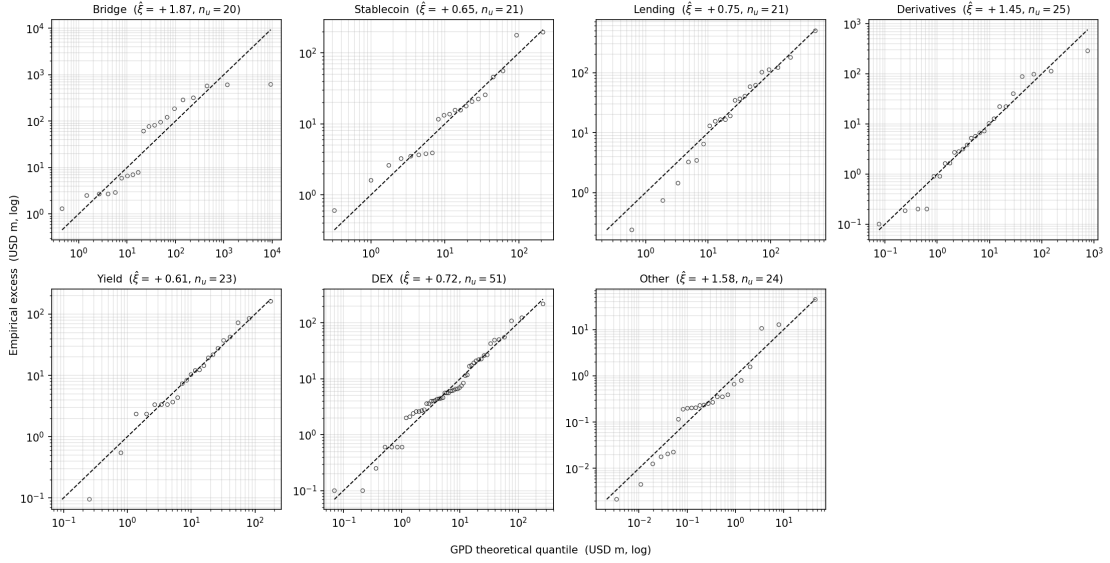


Figure 8: Per-sector POT-GPD QQ diagnostic. Each panel plots empirical excesses $X - u^* \mid X > u^*$ against the fitted-GPD theoretical quantiles at the headline threshold, with the red dashed line $y = x$. A straight line certifies the tail fit at every quantile, the property a 99.9% capital number depends on. The moderate-tail sectors align across the full tail; Lending and Bridge show a slight upper-tail over-shoot (data heavier than the fit), so their figures are mildly conservative.

at $p \leq 0.02$ in every sector but the two smallest, Stablecoin and Derivatives) and the cross-sector Cameron–Trivedi dispersion index $D = 3.27$ ($z \approx 14$), which together certify that the negative binomial, not Poisson, is the right count law.

Robustness, and what each check rules out. Three sensitivity checks bound the ways the headline could be an artifact. *(i) Drop the largest events* asks whether the tail hangs on a single event. Tail-index estimates are known to be sensitive to a handful of extremes (Embrechts–Kluppelberg–Mikosch 1997) and the DeFi fits are no exception: dropping the single largest event moves $\hat{\xi}$ by 0.2–0.4 in most sectors, most sharply in Lending ($\hat{\xi} = 0.75 \rightarrow 0.35$ without Vires Finance). Bridge and DEX are essentially unchanged — their tails are carried by many events rather than one. Under any drop-top rule up to 0.5%, however, every sector retains its below/above infinite-mean characteristic (Lending, Stablecoin, Yield, DEX below $\hat{\xi} = 1$; Bridge, Derivatives, Other above), so the qualitative conclusion is unchanged even as point estimates move. *(ii) Cap sensitivity* asks how much of the capital number is the exposure cap (Assumption 6) rather than the severity fit. Uncapped, Lending rises 18% $\rightarrow$ 22%, DEX 14% $\rightarrow$ 46%, Yield 15% $\rightarrow$ 30%, and Bridge diverges, so the reported figures are cap-bounded floors and should be read as such. *(iii) Confidence level* asks whether the conclusion depends on the stringent 99.9% benchmark. Lowering it to 99.5% and 99% cuts Lending capital 18% $\rightarrow$ 7% $\rightarrow$ 5% and lifts buffered-venue coverage 5% $\rightarrow$ 18% $\rightarrow$ 32%, but even at 99% the buffer covers only about a third of a requirement that is still an order of magnitude above bank operational-risk capital. The finding is therefore not an artifact of the confidence level. A fourth check, the out-of-sample backtest, is discussed under stationarity in Section 8.

7 The two responses and the substitution test

Now I use the yardstick. Every comparison here is an *adequacy* test: observed margin against modeled requirement.

7.1 Response 1: protocol capital buffers (Table 7)

Take the ten largest Lending venues by gross supplied TVL (June 2026). Four disclose an operational-risk reserve (Aave V3, SparkLend, Compound V3, Venus); six hold nothing. Apply the per-protocol allocation of Section 5 to get each venue’s $\mathbb{E}[S]$ and $\text{VaR}_{99.9}$.

The two-layered finding. Against the pure premium $\mathbb{E}[S]$, all four buffers are ample — coverage 159%–599%. Against the tail $\text{VaR}_{99.9}$, all four *collapse* to 2%–9% coverage, a $\sim 95\%$ average shortfall. A buffer sized to the average is nowhere near sized to the tail — precisely because the tail is heavy. Confusing $\mathbb{E}[S]$ with tail capital is the exact error the paper exists to expose.

7.2 Response 2: depositor yield premia (Table 8)

Where the protocol holds no buffer, the residual falls on the depositor, who should be paid a risk premium. I measure it as

$$\text{risk premium} = \underbrace{\text{30-day TVL-weighted stablecoin supply APY}}_{\text{what the depositor actually earns}} - \underbrace{3.70\%}_{\text{3-month T-bill (FRED DTB3)}}.$$

The finding. Across the unbuffered venues the premium averages +54 bps — but the pure premium alone is 64 bps, and the tail-risk premium (per-protocol $\text{VaR}_{99.9}$) is 34–81% of TVL, thousands of bps. So the premium does not even cover the average, let alone the tail. The cross-section is wide, from +257 bps (Fluid) to −79 bps (Kamino), and three venues pay *below* the risk-free rate while exposing depositors to an uncapped tail.

An honesty flag. The spread is a *proxy*, not an identified price. The supply yield also reflects utilization, rate governance, protocol-idiosyncratic risk, and incentives. This is a reduced-form measurement; a controlled panel regression is the natural extension.

Assumption 9 (Yield spread as the risk premium). The excess of the TVL-weighted stablecoin supply APY over the 3-month T-bill is the depositor’s compensation for operational risk. The risk-free benchmark is the T-bill at 3.70%.

Assumption 9 is deliberately the weakest link on the depositor side, and the paper says so. The spread is not identified: it also moves with utilization and incentives, so it over- or under-states the operational-risk premium by an unknown amount. Two features keep the conclusion robust despite this. The gap being measured is an order of magnitude (tens of bps against a $\sim 1,800$ -bps tail), far larger than any plausible mislabeling of the spread, and the benchmark choice is checked: under SOFR, sUSDS, or the Coinbase USD rate the excess yield stays at or below zero. A panel regression that controls for utilization and incentives is the identified-price extension.

7.3 Do the two margins substitute? The Mann–Whitney test

The efficient-market prediction (Section 1) is that buffered venues, having pre-funded the tail, should pay *lower* yield. I test it.

Why a rank test. With only ten venues and non-normal premia with outliers, a t -test is inappropriate. I use a one-sided *Mann–Whitney U test* (Wilcoxon rank-sum), which compares the *ranks* of the two groups’ premia and makes no normality assumption. The one-sided alternative encodes the directional economic hypothesis: buffered < unbuffered.

The result, and its two halves. The buffered median premium is -86 bps against $+39$ bps unbuffered — a 125 bps gap — with one-sided $p = 0.01$. The *direction* is real and significant. But the whole premium cross-section spans just 418 bps, while the tail demands $\sim 18\%$ of TVL. So:

The market prices operational risk *in the right direction* but an order of magnitude too small. Both margins together leave the tail overwhelmingly unfunded.

This one sentence is the paper’s synthesis — the substitution result (direction) and the adequacy result (magnitude) held together.

7.4 From result to policy

Who bears the shortfall. The premium I measure is the posted supply APY — the rate a retail depositor sees and takes. A professional can assess the tail, negotiate bilateral terms, and size or exit; a retail depositor sees only the posted rate and, lacking standardized disclosure, cannot judge whether the yield compensates the risk (Cornelli et al. 2025 on search-for-yield). So the unpriced tail lands disproportionately on the least-equipped party.

Why disclosure, not capital mandates. Because a decentralized protocol may have no responsible legal entity, a bank-style capital mandate is hard to impose. What market discipline requires is *information*, and the LDA supplies its content. The paper recommends a Pillar-3 analogue — standardized loss-and-yield disclosure, buffer-and-tail disclosure, periodic recalibration — and notes the duty can attach to whoever *fronts* access (front-ends, wallets, aggregators, custodial “earn” products), the identifiable points where a retail depositor actually enters.

A pointer on limitations. The obvious referee objections to the two responses — that the buffer–yield association is descriptive rather than causal, that the yield spread is a proxy, that the sample is only ten venues — are limitations of the assumptions behind the tests. Section 8 takes them up together with every other modeling assumption, stating for each the limitation it implies and the fix.

8 Assumptions, the limitations they imply, and how to fix them

Every headline number rests on a chain of modeling assumptions, each stated at its natural point above. This section collects all nine in one place and, for each, names the limitation it implies and the extension that would fix it. It also adds the one limitation that is not a modeling assumption but a property of the ten-venue sample. There are no subsections by design. The aim is a single readable map of where the next paper on this dataset should go.

The assumptions and their fixes at a glance. Table 1 lists the nine assumptions, each with the direction of the error it induces and the extension that would weaken it. Read

the “direction of error” column as the answer to “if this assumption is wrong, which way is the reported capital number biased?” The recurring pattern is worth naming: almost every assumption biases the capital figure *downward* or leaves the tail untouched. Incomplete data lightens only the body, gross losses overstate nothing, ignored contagion understates cluster years, and the earlier window understates the later tail. The reported inadequacy is therefore conservative. Relaxing the assumptions would, on the balance of the signed errors, widen the gap between the risk and its funding rather than close it.

Stationarity is the limitation the backtest quantifies. The frequency and severity laws are fit on the full 2020–2026 window as if stationary (Assumption 5), yet the rolling intensity of Figure 4 shows the process drifting. The out-of-sample backtest puts a number on the cost. Refitting Bridge on the 2020–2023 sub-sample ($n = 37$) yields a degenerate estimate — the plateau selector falls back because $n_u = 11 < 20$ and the fitted $\hat{\xi}$ is not defensible — while the full 2020–2026 sample gives $\hat{\xi} = 1.87$, driven by the Kelp bridge exploit (\$292 m, 2026) that arrived only in the hold-out. Similarly large 2024–2026 events landed in other sectors: Cetus (\$223 m, DEX) and Drift Trade (\$285 m, Derivatives). The lesson is not a specific drift number but the structural one: for the heavy-tail sectors the 2020–2023 sample is simply too small to fit a tail at all, and the full-sample figure is the appropriate forward-looking estimator with the caveat that a single new mega-event materially moves it. The Kupiec coverage p -values are uninformative at a three-year hold-out, so reading the test as “the model is validated” would over-claim. A regime-split fit or a non-stationary background rate is the fix.

One limitation is not a modeling assumption: confounding in the substitution test. The buffer–yield association (Section 7.3) is descriptive, not causal. The buffered venues are also the largest and oldest, so brand and perceived safety may drive the lower yield as much as the buffer, and ten venues cannot separate the two. This does not touch the capital figures, which are fit from the loss data alone, but it bounds the substitution result to a correlation in the predicted direction rather than an identified effect. A larger cross-section or a panel design that controls for venue age and size is the fix.

The fixes, ordered by how much they would move the result. Three extensions would change the numbers materially, and three would sharpen them without overturning the conclusion. A self-exciting frequency model (Assumption 4) is the single most consequential, because cross-protocol contagion is the mechanism most likely to make a bad year much worse than the iid compound allows, and it pushes the tail up, not down. A hierarchical severity model (Assumption 8) would let the largest venues carry heavier tails than the sector collective, which is where the under-reservation is already concentrated. A regime-split or non-stationary fit (Assumption 5) would replace the full-sample point estimate with a point-in-time one, and the backtest suggests the point-in-time tail is currently rising. The remaining three only sharpen. An identified yield price (Assumption 9) via a panel regression would replace the proxy spread with a controlled estimate, but the gap being explained is an order of magnitude, so a cleaner premium would refine the size of the shortfall, not its existence. A recovery-adjusted severity (Assumption 1) would lower every number by about 6%, tightening the conservatism rather than changing the sign. A second-order tail correction (Assumption 3) would trim the finite- u approximation error the diagnostics already show to be small. None of the six reverses the finding, because the finding is a two-orders-of-magnitude gap and every signed error is smaller than that gap.

This companion tracks `paper.tex` / `paper.pdf` as of the current revision. If the paper’s numbers change (re-running `code.py`), the numbers and figures quoted in Section 6 should

be re-checked against output/risk_summary.json.

Assumption	What it buys	Direction of error	Extension that weakens it
Gross loss and reporting completeness (A1)	Uses reported losses directly; no recovery model	Conservative: nets nothing, lightens only the body	Recovery-adjusted (net) severity; censoring model for small losses
Compound structure (A2)	Classical collective-risk form; tail of S from F_X alone	Depends on the two below	Joint severity–frequency and dependence modeling
GPD tail (A3)	Parametric tail, non-parametric body; one shape parameter ξ	Small (u finite); tested by QQ, KS, mean-excess	Higher threshold; second-order (penultimate) tail correction
Event independence (A4)	Draw N once, severities iid; no timing model	Understates cluster years, so VaR is a lower bound	Self-exciting (Hawkes) point process on the same data
Exposure cap (A6)	Finite VaR even for $\xi > 1$ sectors	Load-bearing for Bridge, Derivatives; those rows are floors	Per-protocol exposure paths; contagion-aware cap
Net TVL as exposure base (A7)	Net (idle) TVL: capital as a share of at-risk depositor funds; respects the cap	Approximate for Stablecoin and Other (flagged, not used)	Supply-side net-exposure series per protocol
Per-sector homogeneity (A8)	Per-protocol capital from thin data via TVL share	Understates large/complex venues if risk-per-dollar rises with size	Size covariate or hierarchical (random-effects) severity
Yield spread as premium (A9)	An observable depositor-side compensation measure	Not identified; robust only because the gap is an order of magnitude	Panel regression controlling for utilization and incentives
Stationarity (A5)	One full-sample fit per sector	Earlier window understates the later tail (backtest)	Regime-split fit; non-stationary background rate

Table 1: The nine modeling assumptions, what each buys, the direction in which it biases the reported capital figure, and the extension that would relax it. The recurring feature is that the signed errors point the same way: relaxing the assumptions would, on balance, widen the risk-versus-funding gap the paper reports, not close it.